Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/29/2005
PI Name:	Kalb, Robert M.D.		
Project Title:	Mechanisms of Gravity-evoked Neuronal Plasticity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	19104	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	10/01/2002	**End Date:**	05/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-90		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1582 ; NAG2-951		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Cell and Molecular biological investigations My lab had the privilege of participating in the Neurolab space shuttle mission (STS-90) in May 1998. Among the experiments we performed was a DNA micro-array comparison of spinal cord tissue from flight versus 1G reared control animals. We identified a gene, serum and glucocorticoid-induced kinase (SGK) whose abundance was 4x higher in flight versus control animals. In this proposal we plan a series of ground-based investigations into the molecular and cellular function(s) played by SGK in nervous tissue. Task Objective: 1. Determine if CREB is a direct target of SGK ; 2. Determine the effect of SGK on neurite growth of neurons		

	Task Significance: This is the first gene whose expression is regulated in the CNS (central nervous system). Its role in neuronal function requires definition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Define the role of the SGK serine-theonine kinase in neuronal function.
Task Progress:	Final Report – NASA Grant NAG2-951 (Robert Kalb, MD) My laboratory had the privilege of participating in the NeuroLab Mission (STS-90) in 1998. One of the experiments we performed was a comparative microarray analysis of genes expressed in the spinal cord of flight versus 1G reared developing rats. We identified a panel of transcripts that were upregulated in of flight versus 1G animals as well as a small number of transcripts that were down-regulated in of flight versus 1G animals. The grant proposal entitled "Mechanisms of gravity-evoked neuronal plasticity" (NASA Grant NAG2-951) proposed to study one transcript in detail. The abundance of the mRNA for the serum and glucocorticoid-induced kinase (SGK) showed a 3-4 fold increase in abundance in of flight versus 1G reared animals. An overview of the two papers that were published reporting our characterizations described below. 1. Expression of serum and glucocorticoid-inducible kinase (SGK) is regulated in an experience-dependent manner and can cause dendrite growth. The interaction of an animal with its environment during a critical period in early postnatal life has life-long effects on the structure and function of sensory and motor systems. To gain insight into the molecular mechanisms of experience-dependent development, we challenged young rats to adapt to a new environment that engenders novel motor behavior. Rats born in earth's gravitational field (1G) were subsequently reared for 2 weeks either in the absence of gravity (microgravity) or at 1G. A comparison of gene expression using micro-arrays led to the identification of a panel of differentially regulated transcripts. We report here that the abundance of serum and glucocorticoid-inducible kinase (SGK) is increased in spinal cord tissue from animals reared in microgravity in comparison with 1G-reared controls. The induction of SGK expression can also be achieved by administration of glucocorticoids to animals at 1G or neurons in vitro. Expression of constitutively-active SGK in neurons leads to the elaboration of neuronal dendrites and their branching. Glucocorticoids also lead to dendrite elaboration and this effect can be abrogated by inhibiting SGK activity. Changes in the level of expression of SGK could be part of the mechanism for experience-dependent acquisition of mature neuronal properties. 2. Serum/glucocorticoid-inducible kinase can phosphorylate the cyclic AMP response element binding protein, CREB. To maintain homeostasis, cells often respond to stressful extra-cellular stimuli by new gene expression. Serum/glucocorticoid-induced kinase (SGK) is an immediate early gene whose expression is induced by a variety of extra-cellular stimuli. We examined the possibility that SGK can directly phosphorylate the transcription factor cAMP response element binding protein (CREB). In a cell-free context, SGK physically associates with CREB and SGK phosphorylates it on serine 133. Phospho-serine 133 is essential for stimulating the transcriptional activity of CREB. Further, we show that in a variety of cellular contexts, SGK phosphorylates CREB. Activation of receptor tyrosine kinase pathways or the phosphoinositide-dependent kinase 1 (PDK1) lead to SGK-dependent CREB phosphorylation. Hormonal stimulation of epithelial cells leads to the induction of endogenous SGK and CREB phosphorylation. A dominant negative form of SGK blocks dexamethasone induced CREB phosphorylation. Our studies indicate that stimulation of SGK can lead to CREB phosphorylation suggesting that CREB-dependent gene transcription is an important link between stressful extracellular signals and cellular responses. All papers published with the support of NASA Grant NAG2-951 Inglis, F.M., Crockett, R., Korada, S., Abraham, W.C., ,Hollmann, M., and Kalb, R.G. The AMPA receptor subunit GluR1 regulates dendritic architecture of motor neurons. J. Neurosci. 22:8042-8051 (2002) Stegenga, S. L., Eward, W. and Kalb, R.G. Effects of the "Neurolab" Space Shuttle Mission on Glutamate Receptor Subunit Expression in the Developing Rodent Spinal Cord. In press, Aviat. Space Environ. Med. (2003) 74:615-621 Gazula, V.-R., Roberts, M., Luzzio, C.,. Jawad, A.F. and Kalb, R.G. (2004) The effects of limb exercise after spinal cord injury on motor neuron dendrite structure. J. Comp. Neurol, 476: 130-145 David, S. and Kalb R.G. Serum/glucocorticoid-inducible kinase can phosphorylate the cyclic AMP response element binding protein, CREB (2005) FEBS Lett., 579(6):1534-1538 David, S., Stegenga, S. L., Hu, P., Xiong, G., Kerr, E., Becker, K.B., Venkatapathy, S., Warrington, J.A. and Kalb, R.G. Expression of serum and glucocorticoid-inducible kinase (SGK) is regulated in an experience-dependent manner and can cause dendrite growth. J. Neurosci. (2005) 25:7048-7053
Bibliography Type:	Description: (Last Updated: 08/16/2018)
Articles in Peer-reviewed Journals	David S, Stegenga SL, Hu P, Xiong G, Kerr E, Becker KB, Venkatapathy S, Warrington JA, Kalb RG. "Expression of serum and glucocorticoid-inducible kinase (SGK) is regulated in an experience-dependent manner and can cause dendrite growth." J Neurosci. 2005 Jul 27;25(30):7048-53. PMID: 16049181 , Jul-2005
Articles in Peer-reviewed Journals	David S, Kalb RG. "Serum/glucocorticoid-inducible kinase can phosphorylate the cyclic AMP response element binding protein, CREB." FEBS Lett. 2005 Feb 28;579(6):1534-8. PMID: 15733869 , Feb-2005
Articles in Peer-reviewed Journals	Gazula VR, Roberts M, Luzzio C, Jawad AF, Kalb RG. "Effects of limb exercise after spinal cord injury on motor neuron dendrite structure." J Comp Neurol. 2004 Aug 16;476(2):130-45. PMID: 15248194 , Aug-2004
Articles in Peer-reviewed Journals	Harding SM, Singh NJ, Walton KD. "A sensitive period for the development of motor function in rats: A microgravity study." Gravit Space Res. 2017 Dec;5(2):57-79. http://gravitationalandspacebiology.org/index.php/journal/article/view/767 , Dec-2017